

Intentional Binding Task: Run Guide

v/0.2

Set up the task

Open the project in PsychoPy (the standalone PsychoPy application is recommended). Open and run "Clock test_2025.py" from the project folder.

Before data collection, check config.py. For the intentional-binding task, ensure condition_keys include only IB-singleTone, IB-singlePress, IB-press, and IB-tone, and wildcard_keys are set to ['Self', 'Other']. Confirm these match the intended experimental design to prevent misclassification of conditions.

Keep wildcard_keys = ['Self', 'Other'] to create blocks labelled SELF and OTHER. With these settings, there are 16 experimental blocks (four conditions x two labels x two repetitions).

Check the monitor settings in config.py (monitor distance, width, frame rate, and fullscr). The supplied setup uses 60 Hz and windowed mode; set and test these for the actual display. Test the audio and headphones on the collection computer before participants arrive.

Overview

The goal of the experiment is to test whether there is a difference in intentional binding between actions that are done by the participant (SELF) and passively done by the experimenter (OTHER).

- SELF: The participant presses SPACE at the instructed moment.
- OTHER: The experimenter presses the participant's finger at a random point in time.

In groups of two or three. Assign roles:

- **Experimenter:** give instructions (if no instructor) and provide "passive" movements.
- **Participant:** Perform the task as described and provide clock reports.
- **Instructor** (optional): give instructions and monitor experiment progression.

Procedure

The participants and the experimenter should sit across from each other with their computer between them. The participant must face the screen (to watch the clock) and wear headphones to hear the sound.

In the conditions labelled SELF, the participant must place their index finger on the SPACE key and perform the task (described below) without any interference from the experimenter. In the conditions labelled OTHER, the participant must place their index finger on the SPACE key. The experimenter must place their finger on top of the participant's finger so they can press the participant's finger to press the key.

The participant must closely monitor the clock in all conditions and report the target event in all conditions.

The instructions on the screen are identical in SELF and OTHER conditions, but at the top (above the instructions) it will say if the upcoming block is SELF or OTHER. The instructor/experimenter should be aware of which condition to initiate (the condition order is randomised).

In the OTHER conditions, the experimenter should attempt to press the participant's finger at a random time within 3-10 seconds after the trial starts. Try to vary the waiting times and do not follow any obvious pattern. Avoid giving any verbal or non-verbal cues to the participant.

When the experimenter has pressed the participant's finger, let the participant move their hand freely to report the event (the experimenter must remove their hand). When the participant confirms their report, place your hands to prepare for a new trial.

The built-in instructions for IB-singlePress, IB-press, and IB-tone tell the participant to press. Before running OTHER blocks, explain that the experimenter will press instead and agree on the cue; this overrides the on-screen press instruction. The Self/Other filename label records the intended actor, not the verified actor, so the instructor must follow the protocol consistently and note any deviations.

IB-singleTone has no keypress, so its SELF and OTHER blocks are behaviorally identical tone-only control blocks. Training includes only IB-press and IB-tone and is not labelled SELF/OTHER; explain the actor rule before the experimental blocks.

In tone-only blocks, participants should not press the response key. In the press-plus-tone blocks, the tone follows the press after the 0.25-second delay. The response screen asks for the relevant event time in each block.

The experiment starts with training trials in which the participant makes the press and reports either press or tone (5x each) to familiarise them with the task. The main experiment starts right after this.

When you start the experiment, enter a numeric subject ID in the dialogue (your group number). There are four basic conditions presented in separate blocks: IB-singleTone, IB-singlePress, IB-press, and IB-tone, which are reported in SELF and OTHER versions. There are 15 trials per block, with two repetitions of each block (total $15 \times 4 \times 2 = 120$ trials).

If you need to abort the experiment or start again, press Q or Escape to quit.

Remember to let the participant take short breaks between blocks.

Instructions

Set up the experiment and place the participant and experimenter as described above.

When the task starts, give the following instructions:

In this task, you have to make a self-initiated keypress on the SPACE key, or the person opposite you will press your finger to press the key as a passive movement.

When you have to do the press yourself, press the key whenever you feel the urge to do so. Do not plan or think about when to press in advance – press when you spontaneously feel so.

Each trial starts when the clock starts spinning. Once it starts, you should focus on the fixation mark and relax. Then press the key as you were just told.

Focus your gaze on the fixation mark in the middle of the clock at all times during the experiment. Avoid following the dot with your eyes.

When you press the key, you will sometimes hear a short beep. After a little extra time, the clock will then stop. The task is then for you to indicate the time on the clock when you pressed the key or when you heard the tone. The instructions on the screen will specify whether to report the key press or the tone.

Use the arrow keys to report the time on the clock. Use "LEFT/RIGHT" to move the clock hand a large step. Use "UP/DOWN" to move the clock hand a small step. Confirm by pressing "SPACE". If in doubt, place it as close as possible to the location or as central as possible to where you think you saw it. There are no right or wrong answers in this

task. What we want to measure is your subjective experience of the press, the sound, and the clock.

Once you have confirmed your answer on the clock, a new trial will begin.

In some conditions, there will only be the press (without a tone) or only a tone played without you making a press. In those conditions, you have to report either in the exact same manner. The instructions on the screen will tell you which condition and what to report (I will also give the instructions verbally before each trial).

In the conditions where the person sitting opposite you presses your finger, you relax your hand, keep your gaze fixed at the clock (similar to before) and avoid making any movement with your hand until you have to report the time on the clock.

Any questions?

If everything is clear, start the experiment.

Depending on the condition, you should give the following instructions at the start of each block:

- **IB-press** (press followed by tone; report press): *"Press SPACE as soon as you experience the intention to press // The person opposite you will press your finger. A tone will follow. Report the clock time when you // they pressed."*
- **IB-tone** (press followed by tone; report tone): *"Press SPACE as soon as you experience the intention to press // The person opposite you will press your finger. A tone will follow. Report the clock time when you heard the tone."*
- **IB-singlePress** (press only): *"Press SPACE as soon as you experience the intention to press // The person opposite you will press your finger. Report the clock time when you // they pressed."*
- **IB-singleTone** (tone only): *"Do not press. A tone will occur at a random time. Report the clock time when you heard it."*

If anything goes wrong on a trial (e.g, the participant pressed when they should not or were distracted), press "F". The trial will be marked as an "error", and the experiment will progress to the next trial.

Data and analysis quick start

Output files are in the project's raw_data/ folder. Data are written as CSV files in raw_data, named by subject, condition, label, and repetition (for example,

subject_123_IB-pressSelf1.csv). Use a new subject ID for each run; rerunning an ID will overwrite files with matching names. Training trials are not saved.

To import data, open analysis/import.R in R or RStudio. Set datapath to the project folder, enter the subject IDs in subjects, and keep the four IB condition names and Self/Other wildcard keys.

Run the script; it reads raw_data/ and writes a dated combined CSV to data/.

The script analysis/IB_examples.R has examples on how to get started that you can use.

Then open analysis/IB_examples.R and set datapath to the project's data/ folder. Update infname to the dated combined CSV produced by import.R (the example currently expects combined_cleaned_data.csv).

Set use_condition2 <- TRUE to calculate the example IB effects separately for Self and Other. Run the script after importing. It writes ib_data.csv to R's current working directory, so check that location before running to avoid overwriting a prior result. Review missing-condition warnings and trial counts before interpreting results.

Tasks:

1) Import data

- You can use the script import.R
- Change the "datapath" and/or "raw.path" and the "subjects" variables to match the folder where you put your data and the ID of the research participants (you can leave subjects empty if there are no other files in the folder with the raw data). Make sure the "conditions" and "wildcard_keys" match the experimental conditions.

2) Calculate intentional binding

a) Press binding effect

b) Tone binding effect

- You can get started with the script IB_examples.R
- Change the "datapath" and "infname" to match the folder where you put your data and the combined data file (output from import.R).
- Under the headline Settings, you must state if you want to remove outliers and, if so, what the min/max cutoff in shift time should be. You can also set

the minimum useful trials per condition for calculating intentional binding. Pick values you find appropriate. Document and justify your choices.

- Check and note missing conditions when you import/calculate intentional binding effects. What does missing conditions mean for the further analysis of intentional binding?

3) Plot results (see figures in Moore & Haggard 2008 for inspiration)

- a) Binding effects
- b) Single-trial data (if time)